

# UNIVERSITY OF LONDON

## INTERNATIONAL PROGRAMMES

### BSc Computer Science and Related Subjects



### CM3070 PROJECT

### PRELIMINARY PROJECT REPORT

<Project title>

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| <b>CHAPTER 1: INTRODUCTION</b>      |           |

**Project Template 1:** Deep Learning on a Public Dataset.

**Project Concept:** To predict whether high social media usage affects mental health in youths and young adults.

### **Project Concept**

Mental health is a global pressing issue that many countries are trying to address. According to the 2022 World Mental Health Report by the World Health Organisation (WHO), One in eight people live with mental health disorders (*TEAM, World Mental Health Report: Transforming mental health for all 2022*). Suicide seems to be the most common cause of death among teenagers and young adolescents. What then affects the mental health of young adolescents and teenagers? Well, social media plays a major role in this. This is proven in the 2022 Global Gen Z Survey conducted by McKinsey Health Institute (MHI), where one in four youths responded that their mental health declined over the last three years(*Coe et al., Gen Z mental health: The impact of tech and Social Media 2023*). They were also the group that spent the most amount of time on social media. It is unlikely for youths and young adults to stay away from using social media platforms in this digital world where these platforms and their interactions are in high prevalence. Social media has become part of their lives and also acts as a distressing activity. However, social media usage is proven to be the main cause of mental health disorders among youths according to the 2023 report by the U.S Surgeon General. Hence, our project aims to analyze and predict

whether high social media usage affects mental health in youths and young adults to provide more concrete evidence and back up this claim.

### **Motivations**

Tech companies are constantly making changes to their social media platforms to provide a safer and stress-free environment for youths and young adults to continue to use these platforms more healthily. In 2022, Instagram concealed the number of likes on posts to allow users to feel less pressurized and competitive, in an attempt to help improve the mental health of users. However, this attempt seems to create a lesser impact as users can continue to post comments and are still able to compare themselves with others. In another attempt, Social Media Platforms such as TikTok and Snapchat have a minimum age requirement for users to be at least thirteen years old. However, according to the 2023 report by the U.S. Surgeon General, about 40% of children aged between eight and twelve years old use these platforms. Despite age restrictions and changing platform features, Social media still causes mental health issues in teenagers and young adults as these platforms are built to be addictive and pleasurable. Hence, we need to analyze the time spent by youths and young adults on these platforms and prove that high social media usage affects their mental health.

Key questions to take into consideration:

- Do other factors such as low self-esteem and sleep deprivation affect the mental health of our target audience?
- Are the current available models effective in predicting the claim?
- What are the impacts of the type of platforms on the outcome?

### **Deliverables**

Firstly, a baseline model has to be established after comparing various classification models. The baseline model will set a score for minimum performance and when our model performs better than the score set, then our

model is proven to be efficient. The baseline model is also effective in evaluating other models that might aid us in analyzing the features of the dataset.

Secondly, a machine-learning model has to be built. The model should be able to accurately analyze and classify youths who are at risk of developing mental health issues based on the number of hours they spend using social media platforms. We will inspect how other features affect the intended outcome during the process of building and training the model. This might include the relationship status and even the type of platforms they use. A well-trained model can consider these features and provide an accurate prediction of the results.

Lastly, the model will be evaluated using evaluation metrics. Evaluation metrics will enable us to analyze the effectiveness of the model in predicting the intended outcome. They also allow us to identify the flaws in our model and help us to improve the model by addressing these issues.

## CHAPTER 2: LITERATURE REVIEW

### **A hybrid deep learning approach for depression prediction from user tweets using feature-rich CNN and bi-directional LSTM**

This paper aims to identify depression in Twitter data by proposing a model that is a combination of the Convolution Neural Network(CNN) and a bi-directional Long Short-Term Memory(biLSTM) which achieved an accuracy score of 94.28%(Kour et al., *An hybrid deep learning approach for depression prediction from user tweets using feature-rich CNN and bi-directional LSTM* 2022). The Recurrent Neural Network(RNN) and CNN models are used as baseline models for evaluation. A confusion matrix and AUC curves are also used to evaluate the model.

The biLSTM model is efficient in tackling the vanishing and exploding gradient problems faced by the basic RNN network as it analyses the meaning of a particular word in both directions in a sequence. When the CNN model is combined with the biLSTM model, outstanding results are achieved as the combined model can make use of the bidirectional function of the biLSTM and the feature extraction ability of the CNN model to produce more accurate results. When compared to a classic RNN model, the hybrid model can perform more efficiently as it tackles gradient-related problems to achieve a more precise intended outcome. The biLSTM model can also process longer sequences and CNN uses the backpropagation algorithm to understand and extract important features from the input. Hence, a combination of different deep learning models can interpret datasets more in-depth and more effectively and produce results with a higher accuracy score compared to basic deep learning models such as CNN and RNN. Thus, in this project, a combination of models will be used to evaluate and classify different features in the dataset chosen to build a hybrid model that can detect mental health issues among youths and young adults. Features such as the number of hours spent on social media, the type of platforms the users use, sleep levels, and depression levels will be taken into consideration when building the hybrid model.

## **Big Data Analytics on Social Networks for real-time depression Detection**

The work proves that Random Forest is the most effective machine learning technique in analyzing tweets with demographic features and sentiment analysis for depression in real-time(Angskun *et al.*, *Big Data Analytics on Social Networks for real-time Depression Detection - Journal of Big Data* 2022). The work compares five machine learning techniques, namely Support Vector Machine(SVM), Decision Tree, Naive Bayes, Random Forest, and Deep Learning. The Deep learning technique uses the ReLU activation function with deep feedforward networks that contain a hundred hidden layers(*cite*). The SVM is used as a baseline model for comparison and the models will be evaluated using evaluation metrics such as F-Measure, Accuracy, Precision, and Recall.

The Random Forest and Decision Tree techniques had a higher accuracy score than the baseline model. Still, Random Forest seems to perform better as it is an upgraded model of Decision Tree. The Random Forest Technique provided more precise results as it integrates multiple decision tree outputs to produce a single outcome. This makes it an efficient model that can analyze live Twitter tweets for signs of depression. Random Forest Technique is also easier to use and adapts easily to solve classification and regression problems. Evaluating live tweets and categorizing them accordingly requires the model to be able to examine huge amounts of data at a faster rate, Random Forest seems to be the most appropriate model as it can handle large amounts of data sets skillfully and has high precision levels when predicting the intended outcomes.

However, the downfall of using the Random Forest technique is that more computational resources are needed when using it. It is also an extremely complex model to comprehend due to the vast number of decision trees created to evaluate a large dataset. Random Forest is also set to be slower than the decision tree algorithm and requires a longer time to deduce the results. Hence, the Random Forest technique is not suitable to use in this project as my model needs to be quicker in evaluating the large amounts of data on the time spent by the user on social media platforms, and the type of platforms the users use, and other factors such as lack of sleep of the user causes the development of mental health issues in youths and young adults.

## **Multimodal depression detection on Instagram considering the time interval of posts**

This paper uses a multimodal system to inspect images, text, and behavioral features to predict the depression score of each Instagram post. For analyzing images, the Convolutional Neural Network(CNN) model is used. For the text datatype, a compilation of Recurrent Neural Network(RNN) models such as Long short-term memory (LSTM), Bi-Directional Long short-term (BiLSTM), Gated Recurrent Unit(GRU), and bi-directional Gated Recurrent Unit(BiGRU) are used to build a text classification model. For behavioral evaluation, the Random Forest Classifier is used.

The use of three different models to examine three different types of data is proven to be effective as the model has an F1 score of 0.835. Using the CNN model for image classification is a good choice as it can independently extract features and patterns at a huge scale without having to use feature engineering which makes it an efficient model for processing images. The RNN model is suitable for examining the text data in Instagram posts as there are time intervals present and it is important to consider this aspect when extracting the key features from the text data. RNN models have a memory of history which allows them to capture the background of the input text and evaluate it accordingly(*cite*). A combination of RNN models will allow us to eliminate the vanishing and exploding gradients that basic RNN models experience. The behavioral features are extracted through the Random Forest classifier as it overcomes overfitting and can identify features that are of more importance than others. Hence, using a combination of different deep learning models to examine the different types of data separately and combining the aggression scores to predict and classify users with depression is an efficient method that can be used in this project. The dataset for this project also contains different datatypes such as numerical and text data which can be evaluated separately using Feed-Forward neural networks for numerical data and RNN model for text data. Then, they can be combined using aggression scores which can then be used to predict and classify youths and young adults who are at risk of developing mental health issues.

## **Stress detection using natural language processing and machine learning over social interactions**

This paper aims to examine tweets using Machine learning algorithms such as Random Forest, Decision Tree, and Logistic Regression for sentiment analysis, and the deep learning model Bidirectional Encoder Representations from Transformers (BERT) is used for sentiment classification. The BERT model was able to efficiently identify emotions with an accuracy rate of 94%. Among the three machine learning models used, Random Forest managed to produce the most accurate results with an accuracy of 97.78% (Nijhawan et al., *Stress detection using natural language processing and machine learning over social interactions - Journal of Big Data* 2022). This shows that two different models are used for analysis and classification to enhance the accuracy and precision of the intended outcome.

The BERT model is used for sentiment analysis where a transformer is used to examine the input text data in a bidirectional method which allows it to learn the meaning of a particular word based on its accompanying neighbors. This allows the model to produce a more accurate result compared to other natural language processing models. A Random Forest technique is proven to be the most efficient method in classifying tweets according to their stress levels. As evaluated in the previous sections, Random Forest is effective in providing a more precise result by analyzing large amounts of data but is slower and requires a large amount of resources for computation. In conclusion, we can interpret that it is decisive to use two different models for analysis and classification to achieve results with higher accuracy. However, the dataset I will be using in this project makes use of numerical data and scales mostly apart from the relationship status and the type of social media platforms that are used by users. Hence, a separate model is not required for sentiment analysis and only a combination of classification models can be efficient in detecting mental health issues among youths and young adults.

## **CHAPTER 3:PROJECT DESIGN**

### **Domain and Users**

The scope of the project is focused on the time spent by the young generation on social media platforms and how long hours spent cause the young generation to develop mental health issues. The users of the project include youths and young adults who make up the largest group that spends the most time on these platforms. They are also more vulnerable and prone to developing mental health issues soon.

### **Design Choices**

The dataset I have chosen is obtained from the Kaggle website and is titled “Social Media and Mental Health”. The dataset contains messy textual data that need to be preprocessed and prepared before being used to train and test our model. Data preprocessing methods can be used to handle and clean the data to make it more suitable for training and testing the model.

A hybrid model of a Long Short Term Memory Model (LSTM) and a Recurrent Neural Network will be built to analyze and predict the numerical and textual data types present in the dataset. Then an aggression method will be used to extract the aggression scores of the two models which will then be used to classify and predict users who are more likely to develop mental health issues. The LSTM model can tackle the gradient-related problems faced by the classic RNN model and increase the efficiency and accuracy of the model in general.

## **Overall Structure**

The project will be built on the Jupyter Notebook on my computer. Important libraries will be imported to run the program and build the deep learning model with ease. The dataset will undergo preprocessing steps and will be cleaned before being analyzed and visualized. The dataset will then be converted to the appropriate data structure to be used for building the baseline model. Then, we will build the hybrid model of LSTM and RNN model before performing model evaluation, testing, and prediction. Evaluation metrics such as the confusion matrix will be then used to calculate the accuracy and precision of the model.

## **Software and Methods Used**

Python will be used as the main language to develop the baseline models and the hybrid model due to its availability of an extensive library and packages. It is a user-friendly platform that can deliver projects of great quality. Supervised learning is used to build and train the hybrid model of LSTM and RNN models. The LSTM model is effective in grasping long-term dependencies and can capture important features from input data effectively. as it can analyze both current and past inputs and make accurate predictions. The RNN model can be used to interpret the text data of various lengths which makes it an efficient model in processing different types of textual data in our dataset.

## **Work Plan**

The work plan can be seen in Fig 1.1 and 1.2 in Appendices.

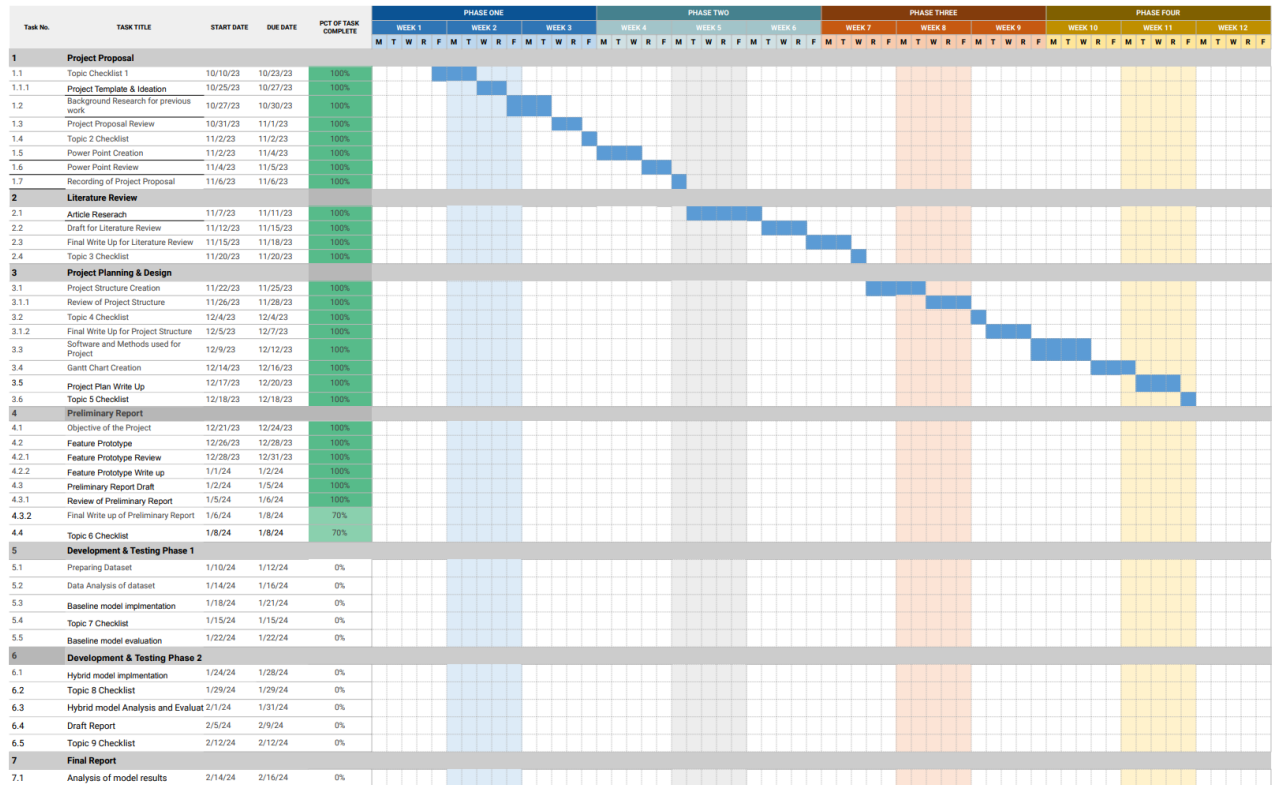
## **Evaluation**

The Evaluation Methods that are going to be used are Balanced Accuracy, F1 Score, and confusion matrix. Since the dataset is set to be imbalanced, it is suitable to use the F1 score as it considers imbalance when evaluating the model. Since the F1 score is an average of both Precision and Recall, it is more effective to use the F1 score to evaluate the model. Balanced Accuracy is used since it considers both sensitivity and specificity when handling imbalanced datasets. The confusion matrix is also used to evaluate the accuracy of the hybrid model.

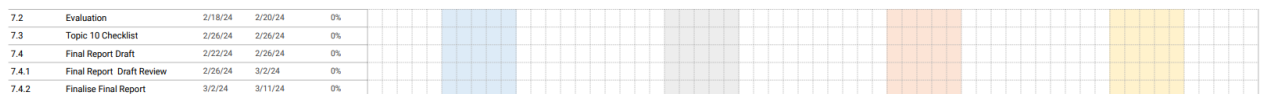
## **CHAPTER 4:FEATURE PROTOTYPE**

The feature we will be evaluating in the prototype is the number of hours spent on social media platforms. We will be using the logistic regression model as a baseline and compare it with commonly used machine learning models such as Random Forest, Decision tree, and Support Vector Machine to evaluate the feature mentioned above. The accuracy score and the confusion matrix are used to evaluate the accuracy and precision of the models in evaluating the feature. Among the three machine learning models, none of the models seem to perform better than the baseline model as they all had an accuracy score of less than 0.5. This shows that the feature selected is not effective in predicting the development of mental health issues among youths and young adults. Hence, another feature is to be selected after performing feature selection.

## CHAPTER 5: APPENDICES



### Fig 1.1 Gantt Chart



### Fig 1.2 Gantt Chart

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